Chapter 13 — The DOM and Events

A web page, as JavaScript sees it, is a tree of objects called the **DOM**. Every interactive feature on the web — a dropdown opening, a todo item ticked off, a form refusing an invalid email — is JavaScript finding nodes in that tree, changing them, and reacting to **events** such as clicks and key presses. This chapter teaches the DOM API (the set of objects and functions JavaScript uses to read and change the page) from the ground up, the event system in full (including the capture/bubble path — the down-then-up route an event travels through the page — that trips up many developers), forms, and the performance rules that keep pages smooth.

What you'll learn

1. What the DOM is: nodes, elements, the tree, and how HTML (the tag-based markup language web pages are written in) becomes a DOM.
2. Selecting elements and walking the tree.
3. Creating, inserting, moving and removing nodes; `textContent` vs `innerHTML`; templates and fragments.
4. Attributes vs properties; `classList`, `dataset` and styles.
5. Events: listeners, the event object, listener options, custom events.
6. Propagation: capture, target and bubble phases; stopping propagation.
7. Event delegation — one listener for many elements.
8. Default actions, `preventDefault`, passive listeners.
9. Forms: `submit`, `FormData`, the Constraint Validation API, `input` vs `change`.
10. Watching the DOM with `MutationObserver`.
11. Rendering performance: reflow, repaint and layout thrashing (measured in Chrome).
12. When scripts run: `DOMContentLoaded`, `load`, `defer`.

Before you start. You'll use closures (Chapter 4) and `this` in handlers (Chapter 9 §9.1). Event listeners run as tasks on the event loop (Chapter 12). The HTML and CSS (the language that styles a page) side — semantic elements (tags chosen for meaning, like `<nav>` or `<button>`), forms markup, selectors (patterns like `.title` or `#app` that pick out elements) — is in the HTML & CSS Handbook (`../../../html-css/`).

How the code runs. Node has no DOM, so the `.mjs` programs in `code/` start with `import '../../../../tools_dom.mjs';`, which creates a **jsdom** document (a JavaScript implementation of the DOM standard used by test tools) and exposes `document`, `window` etc. as globals (variables available everywhere without importing). jsdom implements the DOM and events faithfully but has **no layout engine** (the part of a browser that works out sizes and positions — so no sizes, no painting). For those parts, small `.html` pages were run in real headless Chrome (Chrome running with no visible window, controlled by a script) and screenshotted after a scripted interaction. All outputs are real.

13.1 The DOM tree

What it is

The **Document Object Model (DOM)** is the browser's live, in-memory representation of a page: a **tree** of **nodes**, with an API (defined by the WHATWG — the browser makers' group that maintains the web standards — DOM and HTML standards) for reading and changing it. Change the tree and the browser updates the screen.

Analogy: the HTML file is the architect's blueprint; the DOM is the actual building. JavaScript renovates the building, not the blueprint — "View Source" still shows the original HTML, while DevTools' Elements panel shows the current DOM.

Terms:

- **Node** — any item in the tree. Main types: **element** (`<li>`, nodeType 1), **text** (3), **comment** (8), **document** (9), **document fragment** (11).
- **Element** — a node created from a tag; has attributes and can have children.

- **Parent / child / sibling / ancestor / descendant** — family relations in the tree.
- `document` — the root object; `document.documentElement` is `<html>`, plus `document.head` and `document.body`.

How it works

1. The browser's HTML **parser** (the component that reads text and understands its structure) reads the HTML text and builds the tree, fixing errors in standard ways (unclosed tags, missing `<tbody>`).
2. Whitespace between tags becomes **text nodes** — they exist even though they show nothing.
3. Scripts query and change the tree through the DOM API; the changes are live everywhere.

`code/01_dom_tree.mjs` — key parts:

```
document.body.innerHTML = `
<main id="app">
  <h1 class="title">Groceries</h1>
  <!-- a comment node -->
  <ul id="list">
    <li data-id="1">Milk</li>
    <li data-id="2" class="done">Bread</li>
  </ul>
</main>`;

function show(node, depth = 0) {           // recursive walk over childNodes
  // prints #text / #comment / <tag> ELEMENT with indentation
  for (const child of node.childNodes) show(child, depth + 1);
}
show(document.querySelector('#app'));

const main = document.getElementById('app');
console.log('\nchildNodes (all nodes):', main.childNodes.length, '| children (elements only):',
main.children.length);
// ...
const items = document.getElementById('list').children;   // live HTMLCollection
console.log('items before:', items.length);
document.getElementById('list').insertAdjacentHTML('beforeend', '<li data-id="3">Eggs</li>');
console.log('items after :', items.length);

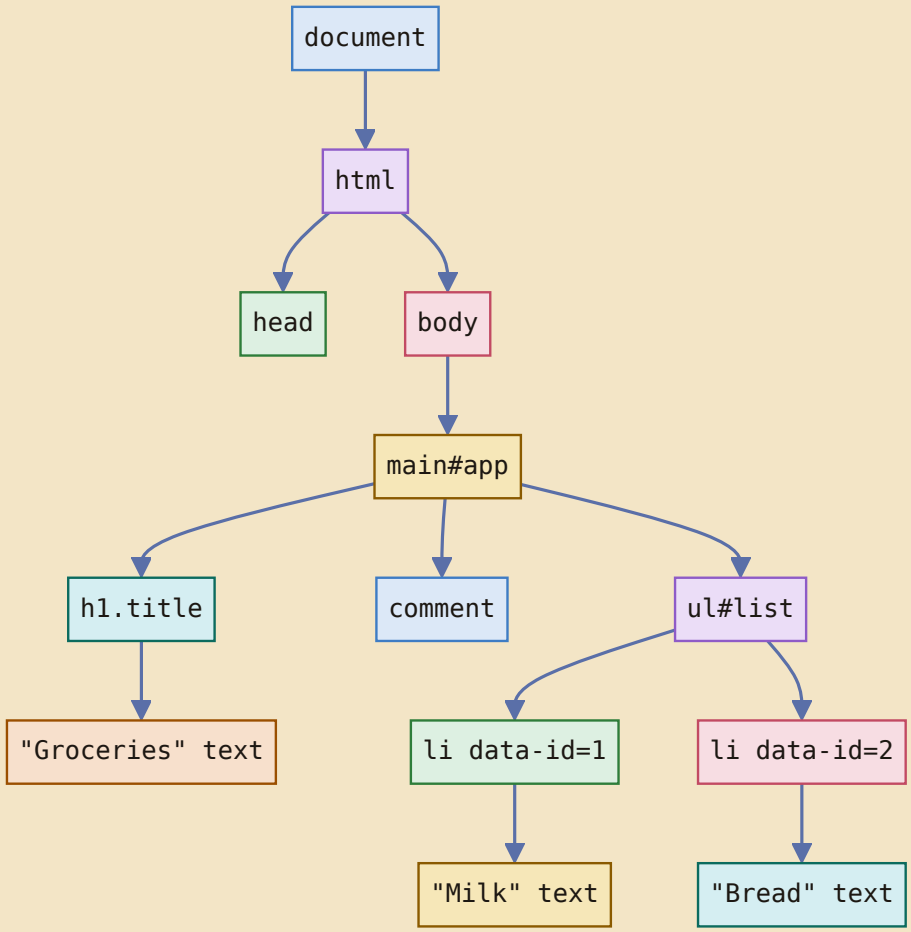
const parsed = new DOMParser().parseFromString('<table><tr><td>cell</table><p>unclosed', 'text/html');
console.log('\nparser auto-inserted <tbody>:', parsed.querySelector('table > tbody > tr') !== null);
console.log('parsed body:', parsed.body.innerHTML);
```

Output:

```
<main> ELEMENT
  #text (whitespace)
  <h1> ELEMENT
    #text "Groceries"
  #text (whitespace)
  #comment "a comment node"
  #text (whitespace)
  <ul> ELEMENT
    #text (whitespace)
    <li> ELEMENT
      #text "Milk"
    #text (whitespace)
    <li> ELEMENT
      #text "Bread"
    #text (whitespace)
  #text (whitespace)

childNodes (all nodes): 7 | children (elements only): 2
document.documentElement: HTML | head: HEAD | body: BODY
nodeType of document: 9 === Node.DOCUMENT_NODE: true
items before: 2
items after : 3

parser auto-inserted <tbody>: true
parsed body: <table><tbody><tr><td>cell</td></tr></tbody></table><p>unclosed</p>
```



The tree for the snippet above (whitespace text nodes omitted).

Common mistakes

- Using `childNodes` / `firstChild` / `nextSibling` and getting whitespace text nodes. Prefer the **element-only** versions: `children`, `firstElementChild`, `nextElementSibling`.
- Assuming the DOM equals the HTML source — the parser inserts, moves and closes elements.

- Running a script in `<head>` that queries `document.body` elements before they're parsed (§13.12).

TL;DR

The DOM is a live tree of nodes (elements, text, comments) built by the parser from HTML. Use element-only navigation properties to skip whitespace text nodes.

13.2 Selecting and traversing

What it is

Method	Returns	Notes
`document.getElementById(id)`	Element or `null`	Fastest; id without `#`
`el.querySelector(css)`	First match or `null`	Any CSS selector; searches descendants of `el`
`el.querySelectorAll(css)`	**Static** `NodeList` (snapshot)	Has `forEach`; spread to an array for `map` / `filter`
`el.getElementsByClassName(c)` / `getElementsByTagName(t)`	**Live** `HTMLCollection`	Updates automatically; no `forEach`
`document.forms.name.elements.field`	Form controls by `name`	See §13.9
`el.closest(css)`	Nearest ancestor **or itself** matching, or `null`	Key to event delegation
`el.matches(css)`	`true` / `false`	"Does this element match?"
`parent.contains(node)`	`true` / `false`	Includes `parent` itself

Traversal properties: `parentElement`, `children`, `firstElementChild`, `lastElementChild`, `previousElementSibling`, `nextElementSibling` (and the node-level `parentNode`, `childNodes`, `firstChild`, `nextSibling`…).

`code/02_select_traverse.mjs` — key parts:

```
console.log('querySelector       :', document.querySelector('.links a').textContent);        // first match
console.log('querySelectorAll    :', [...document.querySelectorAll('.links a')].map((a) => a.textContent));
console.log('attribute selector  :', document.querySelector('[data-track]').textContent);
console.log(':not + :nth-child   :', [...document.querySelectorAll('li:not(:first-child) a')].map((a) =>
a.textContent));
console.log('no match            :', document.querySelector('.nope'), document.querySelectorAll('.nope').length);

const staticList = document.querySelectorAll('li');          // snapshot
const liveList = document.getElementsByTagName('li');         // live
document.querySelector('ul').append(document.createElement('li'));
console.log('\nafter adding an <li>: static', staticList.length, '| live', liveList.length);

const blog = document.querySelector('a[href="/blog"]');
console.log('closest("nav")      :', blog.closest('nav').id);
console.log('matches(":not(.x)") :', blog.matches('a:not(.active)'));
const li = blog.parentElement;
console.log('nextSibling (a node):', JSON.stringify(li.nextSibling.textContent), '<- whitespace text node');

try { document.querySelector('a[href=/]'); } catch (e) { console.log('\nbad selector        :', e.name); }
try { document.querySelector('#123'); } catch (e) { console.log('id starting w/ digit:', e.name, '-> use
getElementById("123") or CSS.escape'); }
```

Output:

```
getElementById      : menu
querySelector       : Home
querySelectorAll    : [ 'Home', 'Blog', 'About' ]
attribute selector  : About
:not + :nth-child   : [ 'Blog', 'About' ]
scoped query        : 3
forms collection    : email
no match            : null 0

after adding an <li>: static 3 | live 4
NodeList has forEach: function | HTMLCollection has forEach: undefined

parentElement       : LI
closest("nav")      : menu
closest(".missing") : null
matches(":not(.x)") : true
previousElementSibling: Home
nextElementSibling  : About
nextSibling (a node): "\n    " <- whitespace text node
firstElementChild   : Home
contains            : true

bad selector        : SyntaxError
id starting w/ digit: SyntaxError -> use getElementById("123") or CSS.escape
```

Common mistakes

- `document.querySelector('.item').classList.add(…)` when nothing matches → `TypeError: Cannot read properties of null`. Check for `null`, or use optional chaining (`?.`, which returns `undefined` instead of throwing when the left side is `null`/`undefined`) when absence is fine.
- Calling `.map` on a `NodeList` — it has `forEach` but not `map`; use `[...list]` or `Array.from(list)`.
- Looping over a **live** collection while removing items from it — indexes shift and items are skipped. Loop over a static copy.
- Unquoted attribute values containing `/` or starting with digits in selectors → `SyntaxError`. Quote them: `a[href="/"]`.

Self-check

1. Which selection method returns a live collection? → `getElementsBy…`. `querySelectorAll` is static.
2. `el.closest('li')` when `el` itself is an `<li>`? → Returns `el`.
3. How do you get the next *element*, skipping text? → `nextElementSibling`.

TL;DR

`querySelector(All)` with CSS selectors for almost everything; `getElementById` for ids; `closest`/`matches`/`contains` for relationships. `querySelectorAll` is a static snapshot; `getElementsBy*` collections are live.

13.3 Creating, inserting and removing nodes

What it is

Task	API
Create	`document.createElement('li')`, `document.createTextNode('hi')`, `document.createDocumentFragment()`
Insert inside	`parent.append(...nodesOrStrings)`, `parent.prepend(...)`, `parent.replaceChildren(...)` (also empties)
Insert beside	`el.before(...)`, `el.after(...)`
Remove / replace	`el.remove()`, `el.replaceWith(...)`
Insert HTML string	`el.insertAdjacentHTML('beforebegin' / 'afterbegin' / 'beforeend' / 'afterend', html)`
Copy	`el.cloneNode(true)` (deep = with all descendants)
Text	`el.textContent = '...'` (safe, never parsed)
Markup	`el.innerHTML = '...'` / `el.outerHTML` (parsed as HTML)

Older equivalents you'll see in legacy code (older code that is still in use): `appendChild`, `insertBefore`, `removeChild`, `replaceChild` (they accept only nodes, not strings).

`code/03_create_modify.mjs` — key parts:

```
const li = document.createElement('li');
li.textContent = 'appended';
list.append(li, 'a text node too');          // append accepts nodes AND strings
// ... prepend, after
list.lastChild.remove();                       // the text node
list.querySelector('li:nth-child(3)').replaceWith(/* new li "replaced" */);
document.getElementById('out').append(list.firstElementChild);   // appending an existing node MOVES it

const userInput = '<img src=x onerror="alert(1)">';
box.textContent = userInput;                    // shown as text, safe
console.log('3) textContent ->', box.innerHTML);
box.innerHTML = '<b>bold</b> and <i>italic</i>'; // parsed as HTML (only for trusted strings!)

p.insertAdjacentHTML('beforebegin', '<span>bb</span>');  // + afterbegin, beforeend, afterend

const frag = document.createDocumentFragment();
for (const name of ['Ada', 'Grace', 'Linus']) {
  const item = document.createElement('li');
  item.textContent = name;
  frag.append(item);
}
list.replaceChildren(frag);                     // one insertion

const tpl = document.getElementById('card');    // <template id="card"><article>...</article></template>
for (const pr of products) {
  const node = tpl.content.cloneNode(true);     // deep copy of the template's content
  node.querySelector('h2').textContent = pr.name;
  node.querySelector('.price').textContent = `$${pr.price}`;
  grid.append(node);
}
```

Output:

```
1) <li>prepended</li><li>after first</li><li>first</li><li>appended</li>a text node too
2) <li>after first</li><li>replaced</li><li>appended</li> | out: <li>prepended</li>
3) textContent -> &lt;img src=x onerror="alert(1)"&gt;
   innerHTML   -> 2 elements; textContent = "bold and italic"
   outerHTML   -> <div id="out"><b>bold</b> and <i>italic</i></div>
4) <span>bb</span><p><span>ab</span>P<span>be</span></p><span>ae</span>
5) <li>Ada</li><li>Grace</li><li>Linus</li> | fragment is now empty: 0
6) <article class="card"><h2>Lamp</h2><p class="price">$25</p></article><article class="card"><h2>Desk</h2><p
class="price">$180</p></article>
7) innerText in jsdom: not implemented (use textContent)
```

`textContent` **vs** `innerText` **vs** `innerHTML`

Property	Reads	Writes	Cost / safety
`textContent`	All text of all descendants, including hidden elements and `<script>` text	Replaces children with one text node	Fast; **safe** for user data
`innerText`	Only *rendered* text (respects CSS `display:none`, line breaks)	Like textContent but `\n` → ` `	Needs layout → slower; not in jsdom
`innerHTML`	Serialized HTML of the children (turned back into a string)	**Parses** the string as HTML and replaces children	**XSS risk** (cross-site scripting — attacker code injected into your page) with untrusted strings; destroys existing child nodes and their listeners

Line 3 of the output shows why `textContent` is safe: the "image" became harmless text (`<img …`). Assigning the same string to `innerHTML` would create a real `<img>` whose `onerror` runs attacker code — the **XSS** attack covered in Chapter 14 §14.10.

Why fragments and templates

- A **`DocumentFragment`** is a lightweight, parentless container. Appending it moves its children in one operation (the fragment ends up empty — see line 5), so the page does one update instead of N.
- A **`<template>`** holds inert markup (not rendered, scripts don't run, images don't load). `template.content.cloneNode(true)` stamps out copies — this is how frameworks-free apps and Web Components (the browser's built-in way to define your own reusable HTML tags) render lists. Fill clones with `textContent`, not string concatenation.

Common mistakes

- `el.innerHTML += '<li>…</li>'` in a loop: re-parses the whole content each time and wipes listeners and input state on existing children. Use `append` or `insertAdjacentHTML('beforeend', …)`.
- Forgetting that appending an existing node **moves** it (line 2). Clone it if you want a copy.
- `cloneNode` doesn't copy listeners added with `addEventListener`.

TL;DR

`createElement` + `textContent` + `append`/`before`/`after`/`remove`/`replaceWith`. Use `textContent` for data, `innerHTML` only for trusted markup. Batch many inserts with a fragment; stamp repeated markup from a `<template>`.

13.4 Attributes, properties, classes, data and styles

What it is

- An **attribute** is what's written in the HTML (`<input value="1">`); read/write with `getAttribute`, `setAttribute`, `hasAttribute`, `removeAttribute`, `toggleAttribute`. Always strings.

- A **property** is a field on the JavaScript element object (`input.value`). Most standard attributes are **reflected** as properties with the same name, but properties can be typed (booleans, numbers), normalized (absolute URLs — the full address including `https://` and the domain) and can hold **live state** that the attribute doesn't.

`code/04_attributes_properties.mjs` — key parts:

```
const qty = document.getElementById('qty');      // <input id="qty" type="number" value="1">
qty.value = '5';                                 // what the user "typed"
console.log('value property       :', qty.value, '| value attribute:', qty.getAttribute('value'), '|
defaultValue:', qty.defaultValue);
console.log('valueAsNumber        :', qty.valueAsNumber, typeof qty.valueAsNumber);
console.log('href property        :', link.href, '(resolved, absolute)');
console.log('href attribute       :', link.getAttribute('href'));

agree.checked = false;                           // <input type="checkbox" checked>
console.log('checked prop / attr :', agree.checked, '/', agree.hasAttribute('checked'));
buy.setAttribute('disabled', 'false');           // still disabled!
console.log('disabled="false"    :', buy.disabled);

buy.classList.add('loading');
buy.classList.remove('primary');
buy.classList.toggle('active');                  // add if missing
buy.classList.toggle('active', false);           // force off
buy.classList.replace('btn', 'button');

console.log('dataset             :', { ...buy.dataset });     // data-product-id="42" data-in-stock="true"
buy.dataset.variantColor = 'blue';               // -> data-variant-color="blue"

buy.style.backgroundColor = 'yellow';            // camelCase property
buy.style.setProperty('--accent', 'teal');       // custom properties need setProperty
console.log('computed color      :', getComputedStyle(buy).color);
```

Output:

```
value property       : 5 | value attribute: 1 | defaultValue: 1
valueAsNumber        : 5 number
href property        : http://localhost/docs?page=2 (resolved, absolute)
href attribute       : /docs?page=2
checked prop / attr : false / true
disabled="false"    : true
after remove         : false
className            : btn primary
classList            : [ 'button', 'loading' ] | contains loading: true
dataset              : { productId: '42', inStock: 'true' }
inStock is a string : string true
new attribute        : blue
style attribute      : color: red; background-color: yellow; --accent: teal;
computed color       : rgb(255, 0, 0)
attributes           : [
  'id',
  'class',
  'data-product-id',
  'data-in-stock',
  'style',
  'data-variant-color'
]
```

Key rules

Rule	Example
For form controls, the **attribute is the initial/default value**; the **property is the current value**	`value` / `defaultValue`, `checked` / `defaultChecked`
Boolean attributes: present = true, whatever the value	`disabled="false"` is still disabled; use `el.disabled = false` or `removeAttribute`
Some names differ	`class` → `className` / `classList`; `for` → `htmlFor`
URL properties are absolute	`a.href`, `img.src`
`data-*` → `el.dataset.camelCase` (words joined, later ones capitalised), always strings	`data-in-stock` → `dataset.inStock === 'true'`
`el.style` = **inline** style only; `getComputedStyle(el)` = final values from all CSS	Computed values are resolved (`rgb(255, 0, 0)`)

Prefer toggling **classes** over setting many inline styles: the look stays in CSS, and one class change is one style recalculation. Inline `style` is best for truly dynamic values (positions, a progress width), ideally through CSS custom properties (CSS variables, named with a leading `--`): `el.style.setProperty('--progress', '40%')`.

Self-check

1. User types "7" into `<input value="1">`. `getAttribute('value')`? → `"1"`; `input.value` is `"7"`.
2. How do you un-disable a button? → `btn.disabled = false` (or `removeAttribute('disabled')`), not `setAttribute('disabled', 'false')`.
3. `data-user-id="9"` → how to read it as a number? → `Number(el.dataset.userId)`.

TL;DR

Attributes = HTML text (strings, initial values); properties = live typed state. Use `classList` for classes, `dataset` for `data-*`, `style` for inline styles and `getComputedStyle` to read final styles.

13.5 Events and listeners

What it is

An **event** is a notification that something happened — a click, a key press, a form submission, a resource finishing loading. An **event listener** (handler) is a function registered to run when a given event type reaches a given **target** (element, `document`, `window`, or any `EventTarget` — the built-in base type for every object that can receive events).

Analogy: a doorbell. `addEventListener('click', fn)` wires a chime to the button; when someone presses it, every wired chime rings in the order it was installed.

How it works

1. `target.addEventListener(type, listener, options?)` registers.
2. The event occurs (user action, network, or `target.dispatchEvent(event)` from code).
3. The browser builds an **event object** and sends it along a path through the tree (§13.6), calling matching listeners with it.
4. User-generated events arrive as **tasks** on the event loop; `dispatchEvent` calls listeners **synchronously**, before it returns.

The event object's most-used members:

Member	Meaning
`type`	`'click'`, `'keydown'`, …
`target`	The element where the event originated (the deepest one clicked)
`currentTarget`	The element whose listener is running now (= `this` in a non-arrow listener)
`eventPhase`	1 capture, 2 at target, 3 bubble
`preventDefault()` / `defaultPrevented`	Cancel the browser's default action (§13.8)
`stopPropagation()` / `stopImmediatePropagation()`	Stop the journey (§13.6)
`isTrusted`	`true` only for real user/browser-generated events
`key`, `code`, `ctrlKey`, `shiftKey`, `altKey`, `metaKey`	Keyboard info (`key` = character/meaning, `code` = physical key)
`clientX/Y`, `button`	Mouse/pointer position and button
`detail`	Data on a `CustomEvent`

`code/05_events_basics.mjs` — key parts:

```
function onClick(event) {
  console.log('1) type:', event.type, '| target:', event.target.id, '| currentTarget:', event.currentTarget.id,
    '| this === currentTarget:', this === event.currentTarget, '| isTrusted:', event.isTrusted);
}
btn.addEventListener('click', onClick);
btn.click();                                    // synthetic click (isTrusted false)

btn.addEventListener('click', () => console.log('2) second listener'));
btn.addEventListener('click', onClick);         // duplicate, ignored
btn.click();

btn.removeEventListener('click', () => {});     // different function: does nothing
btn.removeEventListener('click', onClick);
console.log('3) after remove:'); btn.click();

b2.addEventListener('click', () => console.log('4) once fired'), { once: true });
b2.click(); b2.click();
const ac = new AbortController();
b2.addEventListener('click', () => console.log('   A'), { signal: ac.signal });
b2.addEventListener('click', () => console.log('   B'), { signal: ac.signal });
b2.click(); ac.abort(); b2.click();

b2.onclick = () => console.log('5) handler 1');
b2.onclick = () => console.log('5) handler 2 replaced 1');
b2.click();

input.addEventListener('keydown', (e) => console.log('6) key:', e.key, '| code:', e.code, '| ctrl:', e.ctrlKey));
input.dispatchEvent(new KeyboardEvent('keydown', { key: 's', code: 'KeyS', ctrlKey: true, bubbles: true }));

document.addEventListener('cart:add', (e) => console.log('7) custom event detail:', e.detail));
btn.dispatchEvent(new CustomEvent('cart:add', { detail: { sku: 'A-1', qty: 2 }, bubbles: true }));

console.log('8) before dispatch');
btn.addEventListener('ping', () => console.log('   listener'));
btn.dispatchEvent(new Event('ping'));
console.log('   after dispatch');
```

Output:

```
1) type: click | target: save | currentTarget: save | this === currentTarget: true | isTrusted: false
1) type: click | target: save | currentTarget: save | this === currentTarget: true | isTrusted: false
2) second listener
3) after remove:
2) second listener
4) once fired
   A
   B
   (both removed by one abort)
5) handler 2 replaced 1
6) key: s | code: KeyS | ctrl: true
   input event, value = Ada
7) custom event detail: { sku: 'A-1', qty: 2 }
8) before dispatch
   listener
   after dispatch
```

Listener options

Option	Effect
`capture: true`	Run during the capture phase (§13.6). Must match when removing.
`once: true`	Auto-remove after the first call
`passive: true`	Promise not to call `preventDefault()` — lets the browser scroll without waiting (§13.8)
`signal: abortSignal`	Remove the listener when the signal aborts — great for removing many at once (Ch. 12 §12.11)

`addEventListener` vs `on…` properties vs inline attributes

Style	Multiple handlers?	Notes
`el.addEventListener('click', fn)`	Yes	Options, removable — the standard choice
`el.onclick = fn`	No — assigning replaces (line 5)	Fine for quick scripts
`<button onclick="doIt()">`	—	Mixes JS into HTML, runs code from a string, blocked by a strict CSP (Content Security Policy — a rule list saying which scripts a page may run) (Ch. 14) — avoid

Common mistakes

- `addEventListener('click', handle())` — calls `handle` immediately and registers its return value. Pass the function.
- Removing with a *new* arrow function (line 3's no-op): keep a reference, or use `{ signal }`.
- Arrow function listeners don't get `this = element`; use `event.currentTarget` — it works for both styles.
- Listening for `'onclick'` — the event type has no `on` prefix.

TL;DR

`addEventListener(type, fn, { once, passive, signal, capture })`. The event object tells you `target` (origin), `currentTarget` (listener owner), and lets you cancel defaults or stop propagation. `dispatchEvent` + `CustomEvent` for your own events.

13.6 Propagation: capture, target, bubble

What it is

When you click a button inside nested boxes, you've also clicked each box, the body, the document and the window. The DOM delivers the event along that **propagation path** in three phases:

1. **Capture phase** — from `window` down through each ancestor toward the target; only listeners registered with `{ capture: true }` run.
2. **Target phase** — at the target itself; listeners run in registration order (capture ones first in modern browsers).
3. **Bubble phase** — back up from the target's parent to `window`; normal listeners run. Most events bubble; some don't.

Analogy: a letter delivered down the chain of command (capture), read by the addressee (target), then the reply copied back up to every manager (bubble).

The page `code/propagation.html` registers a capture and a bubble listener on `window`, `document`, `#outer`, `#inner` and the button, then clicks the button once:

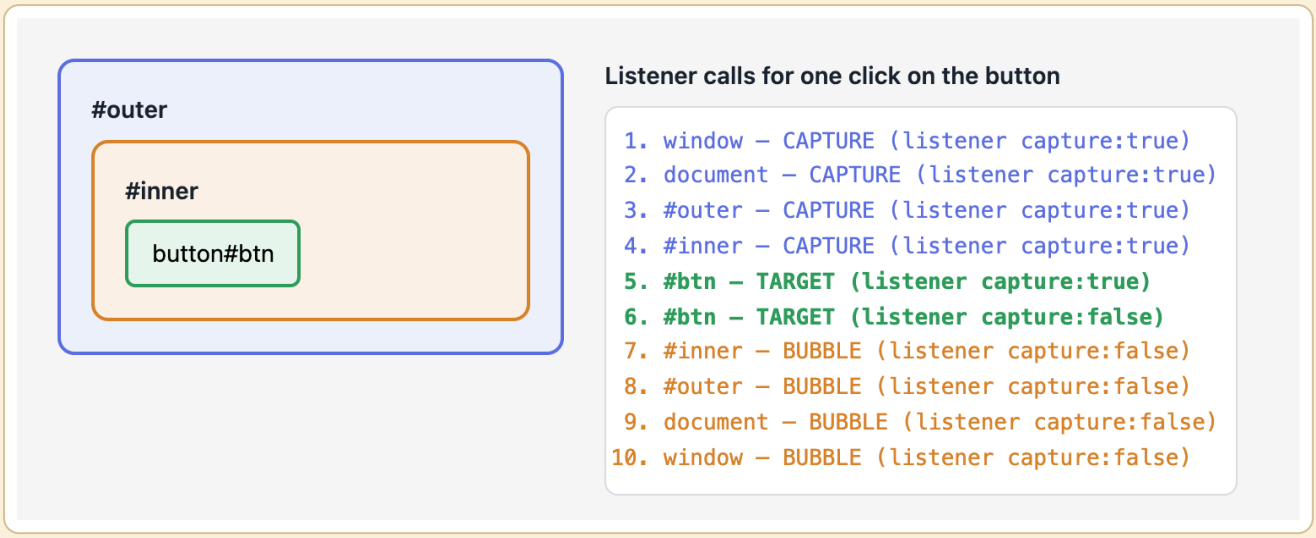


Figure — one click, ten listener calls: capture (blue) goes down, target (green), bubble (orange) goes up. Screenshot from headless Chrome.

`code/06_propagation.mjs` shows the same path in jsdom plus the ways to stop it:

```
for (const el of [window, document, ...els]) {
  const name = el === window ? 'window' : el === document ? 'document' : el.id;
  el.addEventListener('click', (e) => log.push(`${name}:${PHASE[e.eventPhase]}`), { capture: true });
  el.addEventListener('click', (e) => log.push(`${name}:${PHASE[e.eventPhase]}`));
}
document.getElementById('btn').click();

const stop = (e) => e.stopPropagation();
els[1].addEventListener('click', stop);          // on #inner, bubble phase
els[2].click();

btn.addEventListener('click', (e) => { log.push('btn:first-extra'); e.stopImmediatePropagation(); });
btn.addEventListener('click', () => log.push('btn:NEVER'));
btn.click();

document.body.addEventListener('focus', () => seen.push('focus (bubble listener)'));
document.body.addEventListener('focusin', () => seen.push('focusin'));
input.focus();
```

Output:

```
1) full path:
   window:CAPTURE > document:CAPTURE > outer:CAPTURE > inner:CAPTURE > btn:TARGET > btn:TARGET > inner:BUBBLE >
outer:BUBBLE > document:BUBBLE > window:BUBBLE
2) stopPropagation at inner:
   window:CAPTURE > document:CAPTURE > outer:CAPTURE > inner:CAPTURE > btn:TARGET > btn:TARGET > inner:BUBBLE
3) stopImmediatePropagation:
   window:CAPTURE > document:CAPTURE > outer:CAPTURE > inner:CAPTURE > btn:TARGET > btn:TARGET > btn:first-extra
4) new Event() defaults to x: bubbles=false
   body heard: [ 'focusin' ] <- focus does not bubble, focusin does
```

Stopping

Call	Effect
`stopPropagation()`	Finish the listeners on the current element, then stop travelling
`stopImmediatePropagation()`	Also skip the remaining listeners on the current element
`preventDefault()`	Unrelated to propagation: cancels the default action only

Events that don't bubble

`focus` / `blur` (an element gaining / losing **focus**, i.e. becoming / no longer being the one that receives keyboard input; use `focusin` / `focusout` to delegate), `mouseenter` / `mouseleave` (use `mouseover` / `mouseout`), `load` / `error` on elements like `<img>`, `scroll` on elements. Events you create with `new Event(type)` **don't bubble unless you pass** `{ bubbles: true }` (line 4) — a common reason a custom event "isn't heard" by a parent.

When not to stop propagation

`stopPropagation` is a blunt tool: it also blocks analytics listeners (code that records what users do), "click outside to close" handlers on `document`, and framework root listeners (React — a popular UI library — attaches its listeners at the root container). Prefer checking `event.target` in the outer handler, or `defaultPrevented` as a signal.

In the interview

30-second answer: "Events travel in three phases: capture from window down to the target, the target phase, then bubbling back up. `addEventListener` listens in the bubble phase by default; `{capture: true}` switches to capture. `stopPropagation` halts the journey, `stopImmediatePropagation` also skips other listeners on the same element, and `preventDefault` is separate — it only cancels the browser's default action."

- *Q: `target` vs `currentTarget`?* A: `target` is where the event started; `currentTarget` is the element whose listener is running.
- *Q: When would you use capture?* A: To see an event before children can stop it — e.g. a global "close menus on any click" or focus tracking.

TL;DR

Capture (down, `{capture:true}`) → target → bubble (up, default). Most events bubble; `focus`, `mouseenter`, element `load` don't. Stop sparingly.

13.7 Event delegation

What it is

Event delegation means putting **one** listener on a common ancestor instead of one per child, and using `event.target` (usually with `closest`) to find which child was acted on. It works because events bubble.

The problem it solves

- A list of 1,000 rows with 1,000 listeners costs memory and setup time.
- Items added later (from `fetch` — the browser function for network requests — or from the user) would need listeners attached each time — and removed ones should be cleaned up.
- With delegation, new items are handled automatically.

How it works

1. Attach a listener to a stable container (`ul`, `table`, a component root — the outermost element of a UI component — or `document`).
2. In the handler, `const item = event.target.closest('selector')`.
3. If `item` is `null` (click was elsewhere) — or outside the container (`container.contains(item)`) — ignore it.
4. Read what you need from the item (`item.dataset.id`) and act.

`code/07_delegation.mjs`:

```
list.addEventListener('click', (event) => {
  const del = event.target.closest('button.del');
  if (del && list.contains(del)) {
    const li = del.closest('li');
    console.log('delete item', li.dataset.id);
    li.remove();
    return;
  }
  const li = event.target.closest('li');
  if (li) {
    li.classList.toggle('done');
    console.log('toggle item', li.dataset.id, '-> done =', li.classList.contains('done'));
  }
});

let nextId = 3;
document.getElementById('add').addEventListener('click', () => {
  list.insertAdjacentHTML('beforeend', `<li data-id="${nextId}"><span>New task ${nextId}</span> <button
class="del">x</button></li>`);
  nextId += 1;
});

list.querySelector('li[data-id="1"] span').click();      // click on the text (a child of li)
document.getElementById('add').click();                   // adds item 3 AFTER the listener was attached
list.querySelector('li[data-id="3"] span').click();      // still handled - no new listener needed
list.querySelector('li[data-id="2"] .del').click();
```

Output:

```
toggle item 1 -> done = true
toggle item 3 -> done = true
delete item 2
remaining: [ '1(done)', '3(done)' ]
listeners attached to items: 0, to the list: 1
```

The same idea as a real page, `code/todo_delegation.html`: one `click` listener on the `<ul>`, one `submit` listener on the form (with `preventDefault`), and new items built with `textContent` — note the task typed as `Call <b>mom</b>` is shown literally, not as bold HTML. The script adds two tasks, ticks two and deletes one, then the page was screenshotted:

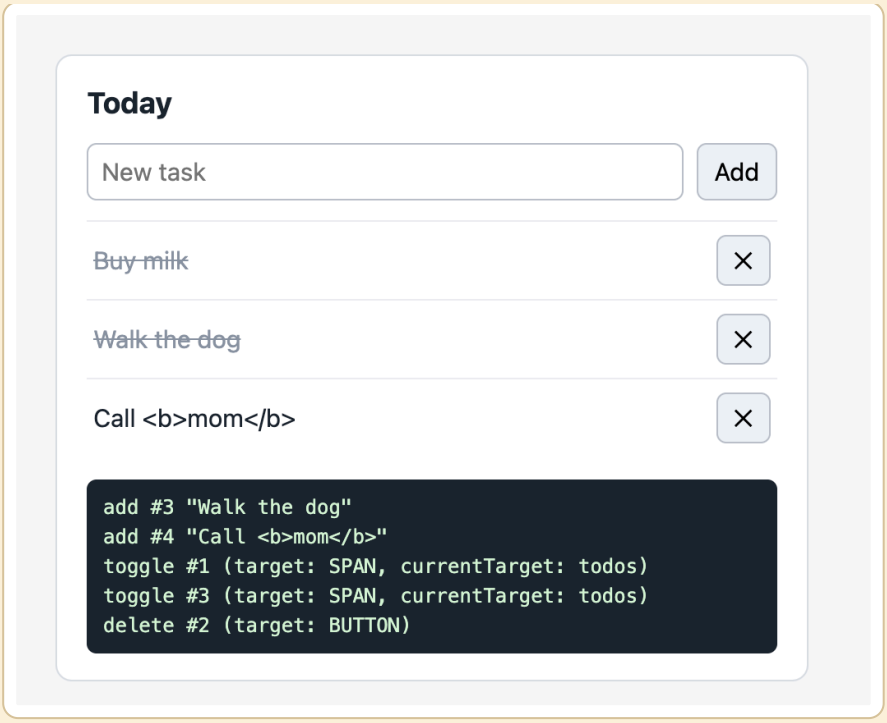


Figure — event delegation in Chrome: `target` is the inner `SPAN` / `BUTTON`, `currentTarget` is always the list.

Why `closest` and not `event.target` directly

`event.target` is the **deepest** element clicked — the `<span>` inside the `<li>`, or an `<svg>` icon inside a button. `closest('li')` climbs to the element you actually care about.

Trade-offs

	Delegation	Listener per element
Listeners	One	N
Dynamic items	Handled automatically	Must attach/detach
Non-bubbling events	Needs the bubbling twin (`focusin`) or capture	Works directly
Handler complexity	Must route by selector	Simple

TL;DR

One listener on the container + `event.target.closest(selector)` handles all current and future children. Use it for lists, tables and menus.

13.8 Default actions and `preventDefault`

What it is

Many events have a **default action** the browser performs after listeners run: a link navigates, a form submits (and reloads the page), a checkbox toggles, a key types a character, a wheel scrolls, a right-click opens the context menu. `event.preventDefault()` cancels it, if the event is **cancelable** (marked as allowed to be cancelled — most user-input events are). `dispatchEvent` returns `false` when a listener cancelled it.

Event	Default action	Typical reason to prevent
`submit` on a form	Send the form and navigate	Validate and send with `fetch` instead
`click` on `<a href>`	Navigate	Client-side routing (switching views and URL without a page reload; History API, Ch. 14)
`keydown`	Insert the character / browser shortcut	Custom shortcut like Ctrl+S
`contextmenu`	Show the browser menu	Custom context menu
`dragover`	Refuse drops	Allow dropping files

(Output lines for `preventDefault` are in the forms program, §13.9.)

Passive listeners

For `touchstart`, `touchmove` and `wheel`, the browser must wait for your listener to finish before scrolling, in case you call `preventDefault()`. That can make scrolling janky (stuttering instead of smooth). `{ passive: true }` promises you won't, so scrolling starts immediately. Browsers other than Safari already default `passive` to `true` for these events when the listener is on `window`, `document` or `body`; calling `preventDefault()` inside a passive listener is ignored (with a console warning).

Common mistakes

- Using `return false` in an `addEventListener` handler — it does nothing there (it only works in `on…` properties).
- Preventing the default of a link or button and forgetting keyboard and screen-reader users (people who use software that reads the page aloud) — use real `<button>`s for actions and `<a>` for navigation.
- `preventDefault()` does **not** stop propagation, and `stopPropagation()` does **not** prevent the default.

TL;DR

`preventDefault()` cancels the browser's built-in response (navigation, submit, typing, scrolling). Use passive listeners for scroll/touch.

13.9 Forms

What it is

Forms are where most user data enters a web app. The DOM gives you:

- `form.elements` — controls by `name` or index; `form.elements.email`.
- `submit` **event** on the form (fires for Enter in a field and for submit buttons) — listen here, not to the button's `click`.
- `FormData` — reads every successful control (named, enabled, checked where relevant) into key/value entries; sends directly as a `fetch` body.
- The **Constraint Validation API** — HTML attributes (`required`, `type="email"`, `min`, `max`, `minlength`, `pattern`) produce a `validity` object; `checkValidity()`, `reportValidity()`, `setCustomValidity(msg)`.
- `input` fires on every change of value; `change` fires when the change is committed (on blur for text, immediately for checkboxes/selects — a select is a dropdown).

`code/08_forms.mjs` — key parts:

```
age.value = '7';
console.log('1) form valid?', form.checkValidity());
for (const el of [email, age]) {
  const v = el.validity;
  console.log(`   ${el.name}: valid=${v.valid} valueMissing=${v.valueMissing} rangeUnderflow=${v.rangeUnderflow}
msg="${el.validationMessage}"`);
}
email.value = 'not-an-email';
console.log('   typeMismatch for "not-an-email":', email.validity.typeMismatch);

email.value = 'ada@example.com';
email.setCustomValidity(email.value.endsWith('@example.com') ? 'Please use your work email' : '');
console.log('2) customError:', email.validity.customError, '|', email.validationMessage);
email.setCustomValidity('');                 // '' clears the custom error

form.addEventListener('submit', (event) => {
  event.preventDefault();                     // stay on the page, send with fetch instead
  const data = new FormData(form);
  console.log('3) submit prevented:', event.defaultPrevented);
  console.log('   entries   :', JSON.stringify([...data.entries()]));
  console.log('   getAll    :', data.getAll('topics'));
  console.log('   as object :', JSON.stringify(Object.fromEntries(data)), '<- keeps only the LAST topic');
  console.log('   submitter :', event.submitter?.textContent);
});
form.elements.topics[1].checked = true;
form.requestSubmit(form.querySelector('button'));  // like clicking the button: validates, fires submit

age.addEventListener('input', () => log.push(`input(${age.value})`));
age.addEventListener('change', () => log.push(`change(${age.value})`));
// ... two input events, then one change event

const a = document.createElement('a'); a.href = '/leave';
a.addEventListener('click', (e) => e.preventDefault());
console.log('5) link click not cancelled?', a.dispatchEvent(new MouseEvent('click', { cancelable: true })));
cb.addEventListener('click', (e) => e.preventDefault());
cb.click();
console.log('   checkbox still checked after prevented click:', cb.checked);
```

Output:

```
1) form valid? false
   email: valid=false valueMissing=true rangeUnderflow=false msg="Constraints not satisfied"
   age: valid=false valueMissing=false rangeUnderflow=true msg="Constraints not satisfied"
   typeMismatch for "not-an-email": true
2) customError: true | Please use your work email
   now valid? true
3) submit prevented: true
   entries   : [["email","ada@example.com"],["age","36"],["topics","js"],["topics","css"],["plan","pro"],
["bio","Hello"]]
   getAll    : [ 'js', 'css' ]
   as object : {"email":"ada@example.com","age":"36","topics":"css","plan":"pro","bio":"Hello"} <- keeps only the
LAST topic
   submitter : Sign up
4) input(4) input(42) change(42)
5) link click not cancelled? false
   checkbox still checked after prevented click: true
```

(jsdom's `validationMessage` is a generic "Constraints not satisfied"; browsers give localized messages — in the user's language — like "Please fill out this field.")

The `validity` flags

Flag	Set when	Caused by
`valueMissing`	Empty required field	`required`
`typeMismatch`	Not a valid email/URL	`type="email"`, `type="url"`
`patternMismatch`	Regex (regular expression, a text pattern) doesn't match whole value	`pattern`
`tooShort` / `tooLong`	Length out of range (after user edits)	`minlength` / `maxlength`
`rangeUnderflow` / `rangeOverflow` / `stepMismatch`	Number/date out of range	`min` / `max` / `step`
`badInput`	Browser can't convert the input (letters in a number field)	input type
`customError`	`setCustomValidity('msg')` was called with a non-empty message	your code
`valid`	None of the above	—

The page `code/form_validation.html` uses `novalidate` (turning off the browser's own bubbles) and renders its own accessible messages (usable by everyone, including people on screen readers) from these flags — `aria-invalid` on the field and `aria-describedby` pointing to the message (ARIA attributes are extra HTML attributes that describe state and relationships to assistive technology). One delegated `input` listener on the form gives live feedback; the `submit` handler checks everything. After a scripted submit with a personal email, age 9 and no password:

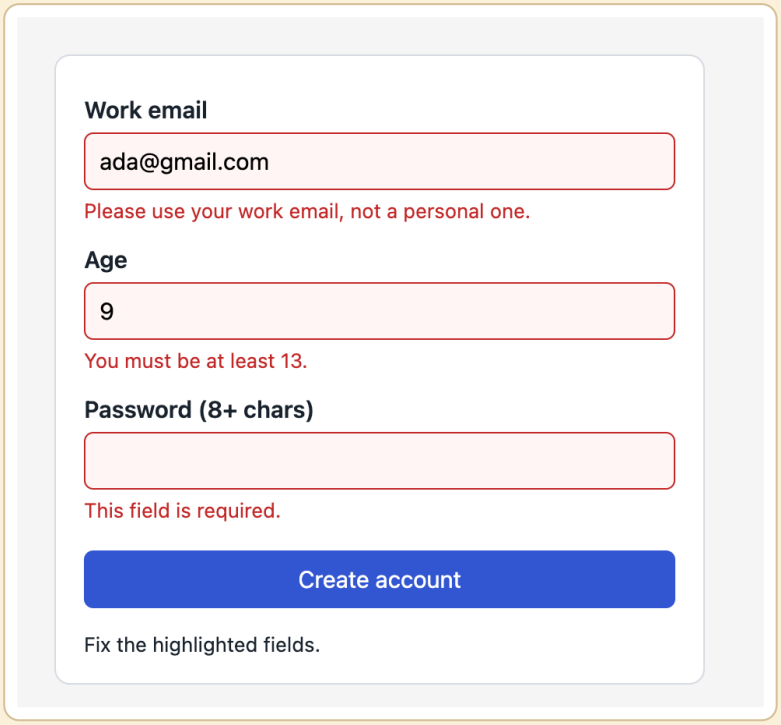


Figure — custom validation messages driven by the Constraint Validation API (headless Chrome).

Common mistakes

- Listening to the submit button's `click` → misses Enter-key submits. Listen to `submit` on the form.
- `Object.fromEntries(new FormData(form))` for multi-value fields (checkbox groups, `<select multiple>`) — keeps only the last value; use `getAll`.
- Client-side validation (checks running in the user's browser) as security. It's UX (user experience) only; **the server must validate everything again**.

- Forgetting `name` attributes — unnamed controls are left out of `FormData`.
- Calling `form.submit()` — it skips validation *and* the `submit` event. `requestSubmit()` behaves like a real submit.

TL;DR

Listen to `submit`, `preventDefault`, read with `FormData`, validate with `validity`/`setCustomValidity` for UX, and validate again on the server. `input` = every keystroke, `change` = committed value.

13.10 Watching the DOM: `MutationObserver`

What it is

`MutationObserver` calls you back when the DOM changes — children added/removed, attributes changed, text edited. Records are **batched** (collected and handed over together) and delivered as a **microtask** (a small queued job the engine runs before handling the next event), after the current script finishes (Ch. 12 §12.2). It replaced the old, slow "mutation events".

`code/09_mutation_observer.mjs`:

```
const observer = new MutationObserver((records) => {
  console.log(`observer got ${records.length} records:`,
    records.map((r) => r.type === 'childList' ? `childList +${r.addedNodes.length} -${r.removedNodes.length}` :
`${r.type}:${r.attributeName}`));
});
observer.observe(feed, { childList: true, attributes: true, subtree: true });

for (let i = 1; i <= 3; i++) feed.insertAdjacentHTML('beforeend', `<li>post ${i}</li>`);
feed.firstElementChild.remove();
feed.setAttribute('data-count', '2');
console.log('sync code done - no callback yet (it is a microtask)');
await null;
observer.disconnect();
```

Output:

```
sync code done - no callback yet (it is a microtask)
observer got 5 records: [
  'childList +1 -0',
  'childList +1 -0',
  'childList +1 -0',
  'childList +0 -1',
  'attributes:data-count'
]
done
```

Uses: reacting to DOM changes made by third-party code (scripts you didn't write, such as ads or chat widgets), auto-initialising widgets (self-contained UI pieces) when they appear, test utilities that wait for elements (Testing Library — a popular UI-testing toolkit — and its `waitFor`/`findBy*` use it). Don't use it to watch your own changes — you already know when you make them. Observers for size and visibility (`ResizeObserver`, `IntersectionObserver`) are in Chapter 14 §14.9.

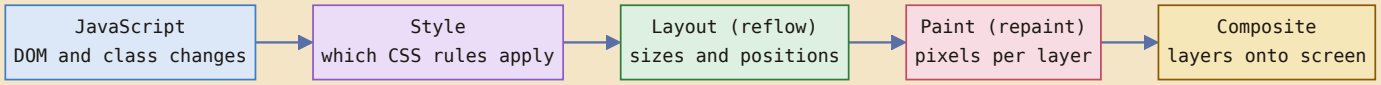
TL;DR

`new MutationObserver(cb).observe(node, { childList, attributes, subtree, characterData })`; batched microtask callbacks; `disconnect()` when done.

13.11 Rendering performance: reflow, repaint and layout thrashing

What it is

After DOM or style changes, the browser's **rendering pipeline** works out what to draw:



- **Reflow / layout** — computing geometry. Changing width, height, font size, adding elements, etc. dirties layout (marks it as needing recalculation) — often for the whole subtree (an element plus everything inside it) or document.
- **Repaint** — redrawing pixels without geometry changes (color, background, shadow).
- **Composite** — assembling layers (separately drawn sheets of the page that the GPU, the graphics chip, stacks together); changing only `transform` and `opacity` can often skip layout and paint entirely, which is why they're the go-to properties for animation.

Normally the browser **batches** all your changes and runs the pipeline once, just before the next frame (the next screen refresh, usually 60 per second; §12.2 step 3). But if you **read** a geometry property (`offsetWidth`, `offsetTop`, `clientHeight`, `scrollTop`, `getBoundingClientRect()`, `getComputedStyle(...)` values) after a write, the browser must run layout **synchronously** to give you a correct answer — a **forced synchronous layout**. Doing write → read → write → read in a loop forces a layout on every iteration: **layout thrashing**.

Measured

The page `code/reflow.html` widens 2,000 boxes by 10 px two ways, timing the median (middle value when sorted) of 7 runs in headless Chrome:

```
// Interleaved (thrashing): each read forces a layout because of the previous write
for (const b of boxes) { const w = b.offsetWidth; b.style.width = (w + 10) + 'px'; }

// Batched: all reads (one layout), then all writes (one layout before the next frame)
const ws = boxes.map((b) => b.offsetWidth);
boxes.forEach((b, i) => { b.style.width = (ws[i] + 10) + 'px'; });
```

Resize 2,000 boxes to 10 px wider than they are (median of 7 runs)

Strategy	Pattern	Time
Interleaved (thrashing)	`for each box: read offsetWidth, then write style.width`	346.3 ms
Batched reads, then writes	`widths = boxes.map(read); boxes.forEach(write)`	1.5 ms

Interleaving forced 2000 synchronous layouts; batching needed one. Here that was about 231x slower (Chrome, this machine - your numbers will differ). First box: 110px (was 100px).

Figure — the same result, two orders of magnitude (roughly 100×) apart (one run in headless Chrome on the author's machine; exact numbers vary, the gap doesn't).

How to keep rendering fast

1. **Batch reads, then writes.** Never interleave them in loops. Libraries like fastdom formalize this.
2. **Change classes, not many individual styles**; let CSS do the rest.
3. **Build off-DOM** (fragment/template) and insert once (§13.3).
4. **Animate `transform` and `opacity`** instead of `top`/`left`/`width`, and use `requestAnimationFrame` for JS-driven animation (Ch. 14 §14.5).

5. **Debounce/throttle** (run only after events stop / at most once per interval) high-frequency handlers (`scroll`, `resize`, `input`) and use passive listeners.
6. **Virtualize** very long lists (render only visible rows) and consider CSS `content-visibility: auto` for long off-screen sections.
7. **Measure** with the DevTools Performance panel — purple "Layout" blocks with a "Forced reflow" warning point straight at the culprit.

In the interview

> **30-second answer:** "The browser recalculates style, layout, paint and composite. It batches DOM writes until the next frame, but reading geometry like `offsetWidth` right after a write forces a synchronous layout. Doing that in a loop is layout thrashing. The fix is to batch all reads, then all writes, animate with `transform` / `opacity`, and use `requestAnimationFrame`."

- *Q: Reflow vs repaint?* A: Reflow recomputes geometry (expensive, cascades); repaint redraws pixels without geometry changes.
- *Q: Which properties avoid layout?* A: `transform` and `opacity` typically only need compositing.

TL;DR

Writes are cheap until you read layout. Batch reads then writes; animate `transform`/`opacity`; measure in DevTools.

13.12 When scripts run: `DOMContentLoaded`, `load` and `defer`

A script can only find elements that have already been parsed. The page lifecycle:

Moment	Event	Meaning
HTML fully parsed, DOM ready, deferred scripts run	`DOMContentLoaded` on `document`	Safe to query any element
Everything loaded (images, styles, iframes — pages embedded in the page)	`load` on `window`	Needed only for image sizes etc.
User leaves / tab hidden	`pagehide`, `visibilitychange`	Save state; `beforeunload` only to warn about unsaved changes

How `<script>` loading affects this:

Tag	Blocks parsing?	Runs when	Order
`<script src>` in `<head>`	Yes (download + run)	Immediately	Document order
`<script defer src>`	No	After parsing, before `DOMContentLoaded`	Document order
`<script async src>`	No (pauses only to run)	As soon as downloaded	Unpredictable
`<script type="module">`	No	Deferred by default	Document order

Recommended: put scripts in `<head>` with `defer` (or use `type="module"`, which is deferred automatically — Chapter 10 §10.7). Then the DOM is ready when your code runs, with no `DOMContentLoaded` wrapper needed. Use `async` for independent scripts like analytics.

```
<!-- (HTML — not run) -->
<head>
  <script src="app.js" defer></script>
</head>
```

> **TL;DR**
>
> Use `defer` or modules so scripts run after parsing. `DOMContentLoaded` = DOM ready; `load` = all resources loaded.

Chapter 13 TL;DR

- **DOM** = live tree of nodes (element, text, comment) built by the parser; use element-only navigation (`children`, `nextElementSibling`).
- **Select** with `querySelector(All)` / `getElementById`; relate with `closest`, `matches`, `contains`. `querySelectorAll` is static; `getElementsBy*` are live.
- **Modify** with `createElement`, `append`/`prepend`/`before`/`after`/`replaceWith`/`remove`, `replaceChildren`. `textContent` for data; `innerHTML` only for trusted markup. Fragments and `<template>` for bulk rendering.
- **Attributes vs properties**: attributes are initial HTML strings; properties are live typed state. `classList`, `dataset`, `style`, `getComputedStyle`.
- **Events**: `addEventListener(type, fn, { once, passive, signal, capture })`; `target` vs `currentTarget`; `CustomEvent` + `dispatchEvent` (synchronous).
- **Propagation**: capture ↓, target, bubble ↑; `stopPropagation` sparingly; `focus`/`mouseenter` don't bubble; `new Event` needs `{ bubbles: true }`.
- **Delegation**: one listener on the container + `event.target.closest()`.
- `preventDefault` cancels default actions; passive listeners for scroll/touch.
- **Forms**: listen to `submit`, use `FormData` (`getAll` for multi-values), the `validity` API and `setCustomValidity`; `requestSubmit()` not `submit()`; re-validate on the server.
- **MutationObserver**: batched microtask callbacks for DOM changes.
- **Performance**: batch reads then writes (thrashing measured ~230× slower), animate `transform`/`opacity`, `requestAnimationFrame`.
- **Loading**: `defer`/modules; `DOMContentLoaded` vs `load`.

Next: talking to servers, storing data, and keeping pages safe — Chapter 14: Browser APIs and Security.